



Do personal taxes affect capital structure? Evidence from the 2003 tax cut[☆]



Leming Lin, Mark J. Flannery*

Warrington College of Business Administration, University of Florida, United States

ARTICLE INFO

Article history:

Received 30 March 2012

Received in revised form

22 October 2012

Accepted 30 October 2012

Available online 6 April 2013

JEL classification:

G32

H24

Keywords:

Capital structure

Personal taxes

Individual ownership

ABSTRACT

Because the personal tax treatments of interest and dividend income likely affect the relative cost of debt and equity financing, a sharp change in tax treatment could affect firms' optimal leverage. This paper examines the effect of the 2003 equity income tax cut on firms' debt usage. Because this tax cut affected only individual investors, we can use a difference-in-differences method to identify the effect of personal tax on firms' leverage. Previous research has found that the 2003 tax cut encouraged dividend payouts and reduced the cost of equity, but it provides no link to equilibrium leverage ratios. We estimate that the tax cut causes the affected firms' leverage to decrease by about 5 percentage points. Furthermore, we show that the effects of the tax cut are stronger for firms with lower marginal corporate tax rates and for firms that are not financially constrained, consistent with our theoretical predictions. Overall, we find strong evidence that personal tax is an important determinant of firms' optimal leverage.

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1. Introduction

In the theory of corporate finance, tax-deductible interest payments encourage firms to finance themselves with debt. But Miller (1977) shows how personal taxes also affect the value of interest versus equity payments to firm claimants. The net tax advantage of \$1 of debt payout, instead of \$1 of equity payout, is

$$(1 - \tau_p) - (1 - \tau_c)(1 - \tau_e) \quad (1)$$

where τ_p is the personal tax rate on interest income, τ_c is the corporate tax rate, and τ_e is the personal tax rate on equity income. The tax benefit of debt increases with the corporate tax rate and the personal tax rate on equity

income, but it decreases with the personal tax rate on interest income. In the extreme, the personal tax rate on interest income (relative to that on equity income) could be large enough to completely offset the corporate tax advantage of debt.

Although important in theory, the empirical evidence that personal taxes affect a firm's leverage ratio is limited. The major difficulty for the empirical tests appears to be identifying variation in personal tax rates that is exogenous and also allows for separating the effects of contemporaneous events. In this paper, we study firms' leverage adjustments following the Jobs and Growth Tax Relief Reconciliation Act of 2003 (JGTRRA). The JGTRRA reduced US federal dividend tax rates from a maximum of 38.6% to 15% and long-term capital gains tax rates from a maximum of 20% to 15% for individual investors. The JGTRRA was passed by the US Congress on May 23, 2003 and signed into law by President George W. Bush on May 28, 2003. The 2003 tax cut provides an ideal setting for testing the impact of personal tax rates on firms' leverage for several reasons. First, the 2003 tax cut introduced a relatively

[☆] We are grateful to the referee (Mike Faulkender), Chunrong Ai, Mara Faccio, Joel Houston, Wayne Losano, Hamed Mahmudi, and Jay Ritter, as well as seminar participants at the 2011 Financial Management Association meetings for helpful comments. All errors are our responsibility.

* Corresponding author. Tel.: +1 352 392 3184.

E-mail address: flannery@ufl.edu (M.J. Flannery).

large increase in the after-tax value of equity distributions. Second, the tax cut came largely as a surprise to the market, so that we can treat it as an exogenous event. Third, the tax cut affected the value of dividends or capital gains only to tax-paying individuals but did not change tax rates for corporations or nonprofit investors. We can, therefore, use the likelihood that each firm's marginal investor is an individual investor to design a difference-in-differences (DID) method of estimating the effect of the 2003 tax cut on firms' leverage.

To motivate our empirical tests, we present a simple model of leverage choice and then conduct a comparative statics analysis of how personal equity income tax affects firms' optimal debt ratios. The model requires knowledge of the benefits and costs of debt. For the benefits, we simply use a modified version of Eq. (1). For the costs, we adopt the reduced form cost function presented in van Binsbergen, Graham, and Yang (2010). The model we present implies that the impact of the 2003 tax cut should be larger for firms with lower marginal corporate tax rates and firms with smaller slopes of the marginal cost function of debt.

To estimate the effect of the tax cut on firms' debt ratio, we examine leverage changes between the ends of fiscal years 2002 and 2004. To eliminate the influence of any confounding factors and to identify the causal effect of the 2003 tax cut, we use percentage of shares owned by individual investors as our main identification variable. We find that a firm whose marginal owner was an individual reduced its book leverage by a considerable amount in response to the tax cut—about 5 percentage points compared with a sample average of 19.9% debt ratio in 2002. This result is consistent with findings that the JGTRRA reduced firms' cost of equity (Dhaliwal, Krull, and Li, 2007) and led them to pay out more in equity income (Chetty and Saez, 2005). Furthermore, the net reduction in affected firms' leverage means that they undertook leverage-decreasing transactions more than sufficient to offset the JGTRRA-induced increase in dividend payments, which would have raised leverage, *ceteris paribus*.¹ Finally, consistent with theoretical predictions, firms with lower marginal corporate tax rates and firms that are less financially constrained reduce leverage ratios more in response to the tax cut.

This paper is related to the literature that deals with the effect of personal taxes on capital structure. Graham (1999) is the first to directly test the effect of personal taxes using firm-level data. He uses dividend payout to proxy for personal tax rates and concludes that adjusting for personal taxes in estimating tax benefits helps to explain debt usage. He also finds that personal taxes are negatively related to debt usage. However, if firms simultaneously select their payout and leverage policies, payout rates could be correlated with leverage, invalidating Graham's interpretation of the payout ratio proxy. Campello (2001) also uses the dividend payout ratio to identify firms that are likely to change their capital structure following the Tax Reform Act of 1986 (TRA). He

finds that zero-dividend firms increased leverage ratios and high dividend payout firms reduced leverage ratios. By examining a period containing an exogenous tax code change, Campello's study could mitigate somewhat, but does not eliminate, the bias from relying on dividend policy for identification. Moreover, since the 1986 act included numerous changes in the relative tax treatment of corporate and individual income, isolating the effect of personal taxes on capital structure is difficult. Dhaliwal, Erickson, and Krull (2007) show that firms are more likely to issue debt (relative to equity) following the tax cuts in 1997 and 2003. However, their analysis of new issues provides only a qualitative assessment of how personal taxes affect equilibrium leverage ratios, and they study only the subset of firms with substantial debt or equity issues. Their methodology thus excludes other mechanisms by which firms could adjust leverage, such as earnings retention. In a contemporaneous study, Faccio and Xu (2011) examine the impact of corporate and personal taxes on capital structure by exploiting the shifts in statutory tax rates across OECD (Organisation for Economic Co-operation and Development) countries from 1981 to 2009. They find that corporate as well as personal taxes are important determinants of capital structure. Although their primary focus is the cross-country differences in the response to tax rate changes, their paper contains a table showing that firms' leverage decreases following the US 2003 tax cut. However, our focus on the US permits more detailed specifications and a fuller assessment of the tax cut, and our estimate is about twice as large as theirs.

Our paper differs from the above studies in several important ways. First, we focus on the relatively simple US 2003 tax cut to test whether personal taxes affect leverage. Second, we adopt a difference-in-differences methodology well suited to quantifying the impact of personal taxes on firms' leverage ratios. Third, we derive our empirical model from a simple theoretical framework, which clearly illustrates how personal equity income taxes impact optimal leverage ratios. Finally, we find evidence consistent with the model's implications, that the personal tax cut reduces firm leverage and this effect decreases with the marginal corporate tax rate and with the slope of the marginal cost function of debt.

The remainder of the article is organized as follows. In Section 2, we review the theory of the tax benefit of debt when personal taxes are considered. After introducing a reduced form cost function of debt, we conduct the comparative static analysis of how personal equity income taxes affect the optimal debt ratio. In Section 3, we discuss the identification strategy and establish our empirical specification. Section 4 describes the data procedure and presents summary statistics. In Section 5, we report our main results about the response of firms' book leverage to the 2003 tax cut. Section 6 provides some robustness checks of the main results, including the use of an alternative variable to measure firms' treatment status. Section 7 concludes the discussion.

2. Theoretical background

In this section, we derive the formula for the marginal effect of the personal equity income tax rate on a firm's

¹ We thank the referee for suggesting this expositional point.

optimal leverage. The derivation draws upon results from Miller (1977), Graham (2003), and van Binsbergen, Graham, and Yang (2010).

We start by following Graham (2003) to present the tax benefit of debt financing. The tax benefit of debt is the tax savings that result from deducting interest from taxable earnings. By deducting a single dollar of interest, a firm reduces its tax liability by τ_c , the marginal corporate tax rate. Assuming that debt is perpetual and that the tax savings are discounted at the debt rate, the present value of the tax benefit of debt is

$$\frac{\tau_c \times rD}{r} = \tau_c D, \quad (2)$$

where r is the interest rate on the debt and D is the amount of debt outstanding. This valuation ignores personal taxes. Although interest payments help firms avoid corporate income tax, interest income is taxed at the personal level at a rate τ_p . Payments to equity holders are taxed at the corporate level (at rate τ_c) and again at the personal level at the personal equity tax rate τ_e .² When personal taxes are considered, the tax advantage of \$1 of debt payout, instead of \$1 of equity payout, is

$$(1 - \tau_p) - (1 - \tau_c)(1 - \tau_e). \quad (3)$$

Discounted at the after-tax interest rate $(1 - \tau_p)r$, the present value of the firm's tax benefit of outstanding debt D is

$$\frac{[(1 - \tau_p) - (1 - \tau_c)(1 - \tau_e)]rD}{(1 - \tau_p)r} = \left[1 - \frac{(1 - \tau_c)(1 - \tau_e)}{1 - \tau_p}\right]D. \quad (4)$$

Thus, the value of a levered firm is equal to the value of the unlevered firm plus the tax benefit of debt

$$V_L = V_U + \left[1 - \frac{(1 - \tau_c)(1 - \tau_e)}{1 - \tau_p}\right]D. \quad (5)$$

The relation (5) indicates that the marginal benefit of debt is constant at $[1 - (1 - \tau_c)(1 - \tau_e)/(1 - \tau_p)]$. Firms, therefore, prefer either all-debt or all-equity financing, according to the magnitudes of τ_c , τ_p , and τ_e . An interior equilibrium results from recognizing the cost of debt. As firms increase debt, they also increase the risk of costly bankruptcy, which leads bond investors to require higher promised bond returns. van Binsbergen, Graham, and Yang (2010) estimate “a linear approximation for ex ante firm-specific [marginal cost] of debt curves” (p. 2111) of the form

$$C' \left(\frac{rD}{B} \right) = \alpha + \beta \frac{rD}{B}, \quad (6)$$

where α is a linear function of firm characteristics including collateralizable assets, total assets, market-to-book ratio, intangible assets, cash flow, and dividend payment.³ In their estimation, C is the annual cost of debt per unit of book assets and B is the book value of assets. Thus, the annual cost of debt is given by $B \times C(rD/B)$. Discounting

the cost by the after-tax interest rate, $(1 - \tau_p)r$, the firm chooses the amount of debt D to maximize

$$V_L = V_U + \left[1 - \frac{(1 - \tau_c)(1 - \tau_e)}{1 - \tau_p}\right]D - \frac{B \times C \left(\frac{rD}{B} \right)}{(1 - \tau_p)r}, \quad (7)$$

whose first-order condition is⁴

$$1 - \frac{(1 - \tau_c)(1 - \tau_e)}{1 - \tau_p} = \frac{B \times C' \left(\frac{rD^*}{B} \right) \times \frac{r}{B}}{(1 - \tau_p)r} \equiv \frac{\alpha + \beta \frac{rD^*}{B}}{1 - \tau_p}. \quad (8)$$

Solving Eq. (8) for the optimal debt ratio gives

$$\frac{D^*}{B} = \frac{1}{\beta r} [(1 - \tau_p) - (1 - \tau_c)(1 - \tau_e) - \alpha]. \quad (9)$$

from which we compute the impact of the equity tax rate on the optimal debt ratio as

$$\frac{\partial D^*}{\partial \tau_e} = \frac{1 - \tau_c}{\beta r}. \quad (10)$$

Given the firm's marginal cost (MC) schedule from Eq. (6), a decline in τ_e shifts the marginal benefit (MB) schedule downward, lowering the firm's optimal debt ratio. The comparative statics result is illustrated in Fig. 1. Eq. (10) implies two things about the size of the leverage change associated with JGTRRA. First, a given cut in τ_e reduces leverage more for firms with lower tax rates τ_c [from Eq. (10), $\partial^2(D^*/B)/\partial \tau_e \partial \tau_c < 0$]. In Fig. 1, the cut in τ_e yields a smaller shift in MB [from Eq. (4)] and thus a smaller change in the optimal debt ratio. Second, a firm's optimal leverage response to a cut in τ_e decreases with the slope of the marginal cost function (β). In Fig. 1, as the slope of the MC line β increases, the same shift in the MB line results in a smaller reduction in the optimal debt level.

To derive an approximate effect of the JGTRRA on firms' leverage, we can apply the van Binsbergen, Graham, and Yang (2010) estimate of β to Eq. (10). They estimate values for β between 7.83 and 17.98 in their Table 4 (for all firms or their Sample A). If $\tau_c = 0.25$ and $r = 0.1$, the marginal effect of equity taxes on the debt ratio is somewhere between 0.42 and 0.96, which implies that the 2003 personal tax cut of 14.3% reduced the typical firm's optimal debt ratio by 5.9–13.7%.⁵ While 5.9% seems plausible, 13.7% is obviously large compared with the average leverage ratio of 19.9% in 2002. In the remainder of the paper, we use a DID method to empirically estimate the effect of JGTRRA on firms' leverage.

3. Identification strategy

Fig. 2 plots Compustat firms' mean book leverage excluding utilities and financial firms from the end of 1998 to the end of 2009. The vertical line denotes the time when the tax cut was passed. There is clearly a sharp

² In practice, the tax rate on dividend income and on capital gains are often different, but our derivation ignores this difference for simplicity.

³ van Binsbergen, Graham, and Yang (2010) note in their footnote 5 that “linearity of the marginal cost of debt implies that the total cost of debt is a quadratic function of interest (x_{it}) [equivalent to our rD]. Further, a positive slope on x_{it} in the marginal cost function implies that the total cost curve is convex.”

⁴ This derivation assumes, for simplicity, that the corporate tax rate does not change with the amount of debt outstanding. Appendix B evaluates the case in which the corporate tax rate declines with outstanding debt and shows that the same qualitative results apply there in nearly all cases.

⁵ The 0.25 is the mean marginal corporate tax rate after interest deductions in our sample and the 10% discount rate is used in, for example, Graham (1996) and Graham and Smith (1999).

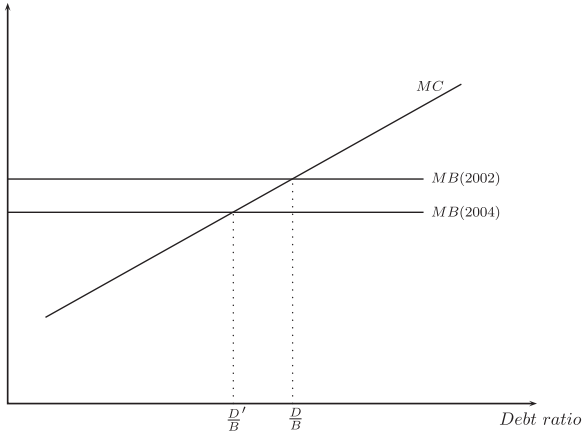


Fig. 1. Marginal cost (MC) and marginal benefit of debt.



Fig. 2. Mean book leverage 1998–2008. Book leverage is defined as the book value of long-term debt (DLT) and short-term debt (DLCC) divided by total assets (AT).

decline in firms' leverage ratios from 2002 to 2004, consistent with the theory that a reduction in τ_e induces firms to cut their debt usage.⁶ However, Fig. 2 does not establish that the tax cut caused lower leverage. This simple comparison of firms' debt levels before and after the tax cut fails to take into account other possible effects. Even though the debt ratio decreased substantially after the tax cut, the drop had started before 2002 and the drop from 2002 to 2004 could be due to other reasons. This is where a difference-in-differences method becomes useful. In this section, we first specify the DID empirical model and testable hypotheses and then discuss the key assumptions that are needed for the DID estimation to yield a consistent estimate of the impact of the 2003 act.

3.1. The DID framework

Eq. (9) implies

$$d\left(\frac{D^*}{B}\right) = -\frac{1}{\beta r} d\alpha + \frac{1-\tau_c}{\beta r} d\tau_e. \quad (11)$$

In van Binsbergen, Graham, and Yang (2010), α is a linear function of firm characteristics X . Assuming that firms are near their optimal leverage ratios, the expected change in the leverage ratio between 2002 and 2004 can be written as

$$E\left(\frac{D^*}{B}_{i,2004} - \frac{D^*}{B}_{i,2002}\right) = \theta \Delta X_i + \left(\frac{1-\tau_{c_i}}{\beta_i r} \Delta \tau_e\right) d_i, \quad (12)$$

where ΔX_i denotes the change from 2002 to 2004 in variables such as collateralizable assets, total assets, market-to-book ratio, intangible assets, cash flow, and dividend payment and d_i is a dummy variable indicating whether firms are affected by the 2003 act: $d_i=1$ if firm i is a treated firm and $d_i=0$ if firm i is a control firm. (We discuss the identification of d_i in Section 3.2.)

Eq. (12) expresses the expected change in leverage ratio as a function of change in firm characteristics and the treatment effect of the tax cut. However, some firm characteristics could impact a firm's leverage through channels that are not captured in our simple trade-off model. For example, the pecking order theory maintains that more profitable firms tend to pay down debt (or retain earnings) and reduce leverage. The market-timing theory predicts that growth firms and firms with strong stock returns are more likely to reduce leverage. Larger firms could find it less expensive to make leverage changes, or they could tend to adjust leverage more quickly. Adding the level of firm characteristics in 2002 ($Z_{i,2002}$) to the explanatory variables in Eq. (12) yields

$$E\left(\frac{D^*}{B}_{i,2004} - \frac{D^*}{B}_{i,2002}\right) = \theta \Delta X_i + \gamma Z_{i,2002} + \left(\frac{1-\tau_{c_i}}{\beta_i r} \Delta \tau_e\right) d_i, \quad (13)$$

where $Z_{i,2002}$ includes end-of-2002 values for total assets, market-to-book ratio, cash flow, and the buy-and-hold stock returns in 2002.⁷

Denoting the treatment effect $((1-\tau_{c_i})/\beta_i r) \times \Delta \tau_e$ by λ_i , rewrite Eq. (13) as

$$E\left(\frac{D^*}{B}_{i,2004} - \frac{D^*}{B}_{i,2002}\right) = \theta \Delta X_i + \gamma Z_{i,2002} + \lambda_i d_i. \quad (14)$$

λ_i is expected to be negative because the JGTRRA lowered the relative cost of equity. In this paper, we are interested in testing three things: (1) What is the average impact of the 2003 tax cut on treated firms, $\bar{\lambda}_i$? (2) Does λ_i decrease with firm i 's marginal corporate tax rate, τ_{c_i} ? (3) Does λ_i decrease with firm i 's slope of marginal cost function of debt, β_i ?

⁶ We compare firms' leverage in 2004 instead of 2003 with that in 2002 because the act was passed in May 2003 and we assume that firms did not adjust to their post-2003 optimal leverage ratios until 2004. In Section 6, we show that a partial adjustment model yields similar conclusions.

⁷ Some previous studies also use the level of firm characteristics to predict the change in leverage (e.g., Givoly, Hayn, Ofer, and Sarig, 1992). Another reason to control for these variables is that our proxy for d_i (individual ownership) is known to be correlated with some of them (Gompers and Metrick, 2001). Omitting them from the regression could, therefore, bias the estimated treatment effect of the 2003 tax cut.

From Eq. (14), we can estimate the average treatment effect of the 2003 tax cut by

$$\frac{D^*}{B_{i,2004}} - \frac{D^*}{B_{i,2002}} = \theta \Delta X_i + \gamma Z_{i,2002} + \lambda d_i + \epsilon_i, \quad (15)$$

where $\lambda = \bar{\lambda}_i$ provides an estimate of the average impact of the 2003 act on the leverage ratios of treated firms.

To test whether λ_i decreases with τ_{ci} as shown in Eq. (13), we add an interaction between the treated dummy and the corporate tax rate,

$$\frac{D^*}{B_{i,2004}} - \frac{D^*}{B_{i,2002}} = \theta \Delta X_i + \gamma Z_{i,2002} + \delta_1 d_i + \delta_2 (\tau_{ci} \times d_i) + \epsilon_i, \quad (16)$$

in which δ_1 is expected to be negative and δ_2 is expected to be positive.

The last task is to test whether λ_i decreases with β_i . Unfortunately, the value of β_i for each firm is unknown. So for this test we rely on the fact that financially constrained firms generally face higher costs of adjusting their debt and, thus, have higher β s than unconstrained firms. Therefore, we estimate Model (15) and Model (16) for financially unconstrained and constrained firms separately. The estimated λ and δ s are predicted to be smaller for financially constrained firms.

3.2. Identifying affected firms

Traditional corporate finance theory suggests that a firm's share price or cost of equity is determined by the firm's marginal investor (e.g., Bell and Jenkinson, 2002). The 2003 tax cut applied only to the equity income distributed to individual investors. Many equity holders were unaffected by the 2003 tax cut, including corporations, tax-exempt universities, and individuals who hold their equities in a tax sheltered retirement account. Therefore, if a firm's marginal investor is an unaffected institutional investor when the 2003 tax act was passed, this firm's cost of equity and optimal leverage would remain the same. However, if a firm's marginal investor is an individual investor, this firm is a treated firm ($d_i=1$) and finds it attractive to reduce the debt in its capital structure. Identification of the difference-in-differences model depends on firms with different ownership structures being differently affected by the JGTRRA. That is, we need to identify firms for which the marginal investor is (more likely to be) a tax-paying individual.

We use two alternative proxies: (1) fraction of a firm's shares owned by individual investors at the time of the tax cut and (2) firms' abnormal stock returns in two event windows when the market received positive signals about the passage of the 2003 tax act.⁸ We use individual ownership as our main proxy and present the results using the abnormal returns as the identification variable in Section 6. Next we make three assumptions that allow us to obtain unbiased estimates using individual ownership as the identification variable. The first and most important assumption is that the probability of a firm being affected by the 2003 tax cut is equal to the fraction

of shares owned by individual investors (*IndOwn*) when the tax cut was passed.

Assumption 1. $Prob(d_i = 1)$ = fraction of shares owned by individual investors at the time of 2003 tax cut (*IndOwn_i*).

Assumption 1 is equivalent to assuming that the probability of a firm's marginal investor being an individual is equal to the percentage of shares owned by individual investors. This assumption is implicitly made in several recent papers (Dhaliwal, Erickson, and Krull, 2007; Brown, Liang, and Weisbenner, 2007; Dhaliwal, Krull, and Li, 2007; Blouin, Raedy, and Shackelford, 2011). **Assumption 1** implies that $E(d_i) = IndOwn_i$. Therefore, we can write $d_i = IndOwn_i + u_i$ with $E(u_i) = 0$. Given this assumption and the two technical assumptions below, estimating Eqs. (15) and (16) by replacing d_i with *IndOwn_i* yields unbiased estimates of λ and the δ s (see Wooldridge, 2001, pp. 63–64).

One possible problem with using individual ownership at the time of the tax cut as the identification variable is that individual ownership would not be constant. In fact, however, this seems to be a limited concern. In our sample, the serial correlation of individual ownership is 0.91 across adjacent years, suggesting that firm ownership is persistent in terms of the composition of institutional and individual investors. Moreover, our main results still hold if we replace the individual ownership in 2002 with the average of individual ownership from 2002 to 2004.

Next we present two technical assumptions that are required for *IndOwn_i* to be a valid proxy for d_i (see Wooldridge, 2001, pp. 63–64).

Assumption 2. *IndOwn_i* is redundant or ignorable given d_i . That is,

$$\begin{aligned} E\left(\frac{D^*}{B_{i,2004}} - \frac{D^*}{B_{i,2002}} \middle| \Delta X_i, Z_i, d_i, IndOwn_i\right) \\ = E\left(\frac{D^*}{B_{i,2004}} - \frac{D^*}{B_{i,2002}} \middle| \Delta X_i, Z_i, d_i\right) \end{aligned} \quad (17)$$

This assumption says that *IndOwn* is irrelevant once X , Z , and d have been controlled for. It is similar to assuming that the coefficient of the treatment variable (*IndOwn_i*) is zero in the absence of the intervention (Angrist and Krueger, 1999). As a check for this assumption, we repeat the analysis in years when there were no changes in tax rates and show that the coefficients on *IndOwn* are not distinguishable from zero.

Assumption 3. $Cov(u_i, \Delta X_i) = 0$ and $Cov(u_i, Z_i) = 0$, where u_i is equal to $d_i - IndOwn_i$.

Assumption 3 simply says that firms' characteristics X_i and Z_i are not determinants of the treatment status of firms in 2003. No obvious reason exists to expect X_i or Z_i to affect the identity of a firm's marginal investor given the firm's ownership *IndOwn_i*.

4. Sample overview

We obtain quarterly data on institutional ownership from the Thomson-Reuters Institutional Holdings (13F)

⁸ We thank Mara Faccio for suggesting this approach.

Database from 1998 to 2009.⁹ This database was formerly known as the CDA-Spectrum database and contains ownership information by institutional managers with \$100 million or more in securities under management. We then match the most recent quarterly ownership data before the fiscal year end with the Compustat Fundamentals Annual database for financial statements.¹⁰ We exclude financial firms [standard industrial classification (SIC) 6000–6999] and utilities (SIC 4900–4999), firms with missing or negative value of debt or assets, and firms with book leverage less than zero or greater than one. We also require that firms have consecutive ownership and leverage data from 2002 to 2004. These procedures leave 34,158 firm years, of which 29,608 have available marginal corporate tax rates data, as estimated by Blouin, Core, and Guay (2010) [downloaded from Wharton Research Data Services (WRDS)]. Our main results rely on a sample of 3,051 firms that have necessary data available in 2002 and 2004. As described in Appendix A, we winsorize several variables. Table 1 defines the variables used in the study and reports summary statistics.¹¹

The mean (median) book leverage is 19% (15%).¹² In our sample, 51% of shares are owned directly by individual investors. Our individual ownership is a bit larger than that reported in Dhaliwal, Krull, and Li (2007). In their Table 2, the mean (median) institutional ownership of 0.61 (0.66) implies that less than 40% of shares are owned by individual investors. There are at least two possible reasons for this difference. First, following Gompers and Metrick (2001), if a stock in Center for Research in Security Prices (CRSP) is not in the 13F database, we set the institutional ownership to zero.¹³ It is not clear if Dhaliwal, Krull, and Li (2007) exclude the stocks that are not held by institutions in 13F. Second, we set institutional ownership to one if it is greater than one.¹⁴ This could also result in a smaller average institutional ownership if Dhaliwal, Krull, and Li (2007) do not make the same adjustment.

A firm's marginal tax rate is the present value of current and expected future tax payment for an additional dollar

earned today. Due to the carryforwards and carrybacks of taxable income in the tax code, a firm's marginal tax rate is determined not only by its current income but also by the income in prior years and its expected income in the future. Researchers have used different methods to estimate the marginal tax rates (Shevlin, 1990; Graham, 1996; Blouin, Core, and Guay, 2010). In this paper we use the estimates by Blouin, Core, and Guay (2010), which differ according to their treatment of interest expenses. The mean (median) marginal corporate tax rate after interest deductions (*Mtra*) is 25% (30%). The mean (median) marginal corporate tax rate before interest deductions (*Mtrb*) is 27% (32%).

5. Debt ratio changes: 2002–2004

In this section, we present the estimation results of the impact of the 2003 tax cut on firms' leverage. We also investigate how the impact varies according to firms' marginal corporate tax rates and financial conditions.

5.1. Univariate results

Fig. 3 presents a graphical illustration of the DID method described in Section 3. For the purpose of this illustration we divide our sample into three equal-size groups based on individual ownership (*IndOwn*) in 2002: firms with *IndOwn* from 0 to 36%, firms with *IndOwn* from 36% to 74%, and firms with *IndOwn* above 74%. The mean leverage ratio of each group from 1998 to 2009 is then plotted in Fig. 3. Leverage falls at all three groups of firms after 2002, but the group with high individual ownership shows a much sharper decline in debt ratios than the other two groups. The most striking feature in Fig. 3 is the comparison of middle and high *IndOwn* groups. Before the tax cut, the two groups of firms have almost exactly the same average leverage ratios. After the tax cut, firms with high *IndOwn* operate with lower leverage ratios. We see no large difference in leverage changes between the middle *IndOwn* and low *IndOwn* firms following the 2003 tax cut.

Ownership is known to be highly correlated with firm size (Gompers and Metrick, 2001). Fig. 4, therefore, plots the time series of firms' leverage by size and individual ownership. We first divide firms into three equal groups based on firm size (book assets), and for each size group we then sort firms into two halves based on ownership. For small-size and middle-size groups, firms with high *IndOwn* exhibit more negative adjustment in leverage than firms with low *IndOwn*. For example, among middle-size firms, those with low *IndOwn* increase their debt ratios by about 1.5 percentage points from 2002 to 2004, while high *IndOwn* firms' debt ratios dropped more than 3 percentage points during the same time. For the large-size firms, the difference in debt ratios between high *IndOwn* firms and low *IndOwn* firms varies little after 2003. Overall, Figs. 3 and 4 show some evidence that individual ownership plays a role in firms' leverage adjustment around 2003. Next we present regression results that control for other variables that could affect firms' leverage.

⁹ Our main results rely on data only from 2002 to 2004. The whole sample is used only for the falsification tests and the partial adjustment model. The sample starts from 1998 because of a concern with the possible confounding impact of the Tax Relief Act of 1997.

¹⁰ We follow the SAS program by Denys Glushkov, Rabih Moussawi, and Luis Palacios from Wharton Research Data Services (WRDS) to extract ownership data from 13F. We thank them for making the program available on WRDS: <http://wrds-web.wharton.upenn.edu/wrds/research/applications/ownership/Institutional%20Ownership/>.

¹¹ Our main results define book leverage as book debt over total assets. We alternatively measure leverage as book debt over debt plus equity and as (debt – cash)/assets, and we get similar estimation results.

¹² The leverage values from 1998 to 2009 were lower than those reported by researchers who use a longer Compustat time series (e.g., Flannery and Rangan, 2006; Lemmon, Roberts, and Zender, 2008). Frank and Goyal (2009) show a decline in both short-term and long-term leverage after the 1980s (see their Table A1). For example, the mean ratio of long-term debt over assets is 0.180 in 1970–1979, and the ratio decreases to 0.116 in 1990–1999 and 0.095 in 2000–2003.

¹³ Omitting firms with missing 13F data has no qualitative effect on our results, e.g., in Table 7.

¹⁴ The shares of a firm owned by institutional investors can be greater than the shares outstanding if there is short selling that expands the gross number of shares available.

Table 1

Summary statistics.

The sample period is from 1998 to 2009. Book leverage (*Blev*) is defined as the sum of long-term debt (Compustat item DLT) and short-term debt (DLCC) divided by total assets (AT). Individual ownership (*IndOwn*) is the fraction of shares owned by individual investors, which is calculated as one minus the fraction of shares owned by institutional investors, obtained from Thomson Reuters Institutional Holdings (13F) Database. $\ln(\text{assets})$ is the logarithm of total assets. Collateralizable assets (*Col*) is the sum of inventories (INVT) and property, plant, and equipment (PPENT) over total assets. Market-to-book ratio (*MB*) is market value of equity (CSHO*PRCC_C) over book value of equity (CEQ). Intangible assets (*Intang*) is intangible assets (INTAN) over total assets. Cash flow (*CF*) is defined as operating income before depreciation (OIBDP) over total assets. Dividend payment (*Ddiv*) is equal to one if common dividends (DVC) is greater than zero, and zero otherwise. *Momentum* is the firm's buy-and-hold return in the 12 months before the end of the fiscal year. *Mtrb* and *Mtra* are firms' marginal corporate tax rates before and after interest deductions estimated by Blouin, Core, and Guay (2010).

Variable	Mean	Median	Standard deviation	Minimum	Maximum	Number of observations
<i>Blev</i>	0.19	0.15	0.19	0.00	1.00	34,158
<i>IndOwn</i>	0.51	0.49	0.31	0.00	1.00	34,158
$\ln(\text{assets})$	5.66	5.58	2.06	−1.95	13.59	34,158
<i>Col</i>	0.37	0.34	0.25	0.00	1.00	33,892
<i>MB</i>	2.86	2.04	2.50	0.10	10.00	34,158
<i>Intang</i>	0.15	0.08	0.18	0.00	0.98	32,010
<i>CF</i>	0.05	0.11	0.23	−1.00	1.00	34,158
<i>Ddiv</i>	0.28	0.00	0.45	0.00	1.00	34,158
<i>Momentum</i>	0.17	0.01	0.89	−1.00	10.00	33,578
<i>Mtra</i>	0.25	0.30	0.11	0.00	0.39	29,608
<i>Mtrb</i>	0.27	0.32	0.10	0.00	0.39	29,608

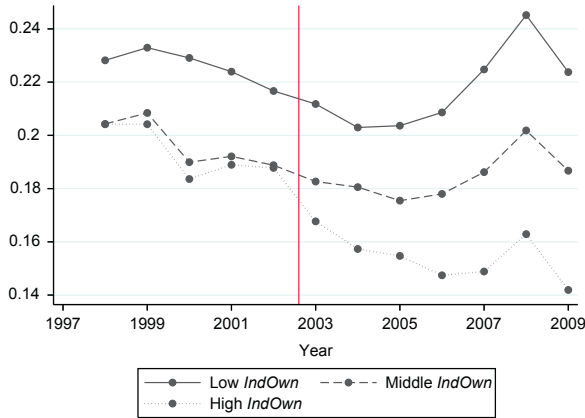


Fig. 3. Mean book leverage 1998–2009 by individual ownership (*IndOwn*) in 2002.

5.2. Multivariate results

Based on the discussion in Section 3, we use individual ownership as a proxy for d_i in Eq. (15) and our regression takes the form

$$\begin{aligned} \Delta Blev = & \alpha + \beta_1 IndOwn + \beta_2 \ln(\text{assets}) + \beta_3 CF \\ & + \beta_4 MB + \beta_5 Momentum + \beta_6 Col \\ & + \beta_7 \Delta \ln(\text{assets}) + \beta_8 \Delta MB + \beta_9 \Delta Intang \\ & + \beta_{10} \Delta CF + \beta_{11} \Delta Ddiv + \epsilon, \end{aligned} \quad (18)$$

where $\Delta Blev$ is the change in book leverage from 2002 to 2004. Thus, the Z_i in Eq. (15) includes end-of-2002 values for total assets, cash flow, market-to-book ratio, and the buy-and-hold stock returns in 2002 (*Momentum*), and X_i includes the same variables as in the cost function of debt in van Binsbergen, Graham, and Yang (2010):

collateralizable assets, total assets, market-to-book ratio, intangible assets, cash flow, and the indicator variable for dividend-paying firms. As discussed in Section 3, β_1 measures the impact of the tax cut on an affected firm's leverage and is expected to be negative.

Column 1 of Table 2 presents the estimation results without including the control variables Z . Consistent with the model's prediction, the coefficient of *IndOwn* is -1.68 and statistically significant. But the estimate could be biased if *IndOwn* is correlated with some of the variables in Z . In the second column, when the control variables Z are included in the regression, the coefficient of *IndOwn* becomes -4.95 , which indicates that the 2003 tax cut causes the treated firms to reduce their leverage by almost 5 percentage points.

The results also show that firms with higher cash flow in 2002 are more likely to reduce their debt ratio, consistent with the pecking order theory. Firms with higher returns (*Momentum*) and higher market-to-book ratios in 2002 reduce their leverage more. This is consistent with a market timing story: Firms tend to issue equity following high stock returns, thus reducing leverage ratios. Because in our sample *IndOwn* is negatively correlated with cash flow, returns, and market-to-book ratio, and all these variables negatively impact the dependent variable, failing to control for these variables underestimate the coefficient of *IndOwn*, exactly as in Table 2.

5.3. Marginal corporate tax rates

Theory [Eq. (10)] indicates that a personal equity income tax cut should have smaller effect on firms with higher marginal corporate tax rates. To test this, we add an interaction between *IndOwn* and *Mtra* to the specification Eq. (18). To avoid multicollinearity among the regressors and to facilitate interpretation of results, we center *IndOwn* and *Mtra*, expressing each variable as deviations from its

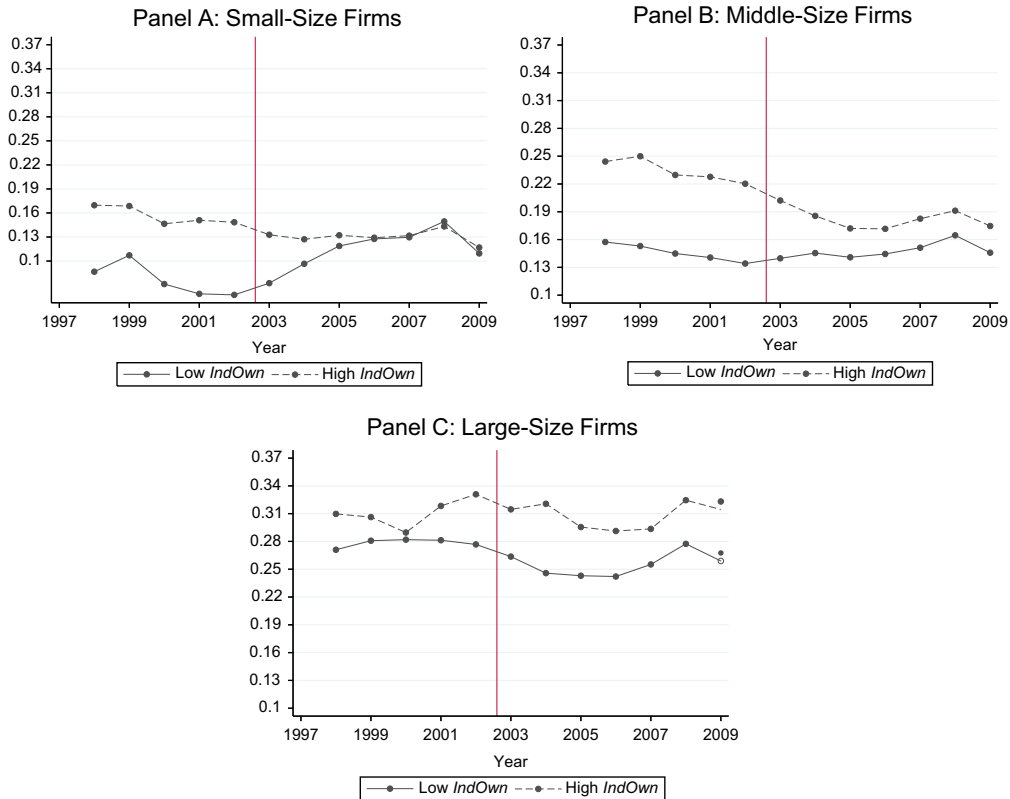


Fig. 4. Mean book leverage 1998–2009 by individual ownership (*IndOwn*) in 2002 for three equal-size categories. (a) Panel A: small-size firms, (b) Panel B: middle-size firms and (c) Panel C: large-size firms.

mean value, and interact the centered *IndOwn* variable with the centered *Mtra* variable.¹⁵

$$\begin{aligned} \Delta Blev = & \alpha + \beta_1 IndOwn_{centered} + \beta_2 IndOwn_{centered} \times Mtra_{centered} \\ & + \beta_3 Mtra_{centered} + \beta_4 \ln(assets) + \beta_5 CF + \beta_6 MB \\ & + \beta_7 Momentum + \beta_8 \Delta Col + \beta_9 \Delta \ln(assets) + \beta_{10} \Delta MB \\ & + \beta_{11} \Delta Intang + \beta_{12} \Delta CF + \beta_{13} \Delta Ddiv + \epsilon \end{aligned} \quad (19)$$

The resulting coefficient of *IndOwn* measures the effect of the 2003 tax cut when *Mtra* is at its mean value in 2002 (20.4%).¹⁶ We also add *Mtra* to the regression to get a clean estimate of the interaction term's coefficient. (The firm's applicable corporate tax rate could also be considered a component of *Z*.)

Column 1 of Table 3 presents the result of the regression without including the control variables *Z*. The regression coefficient of *IndOwn* is now -3.11 , compared with -1.68 in Table 2, when we ignored the effect of marginal corporate tax rates. The interaction term $IndOwn \times Mtra$ carries a significantly positive coefficient, consistent with

the theoretical prediction that firms with higher marginal corporate tax rates reduce their leverage ratio less. After we include the control variables in Column 2, the coefficient of *IndOwn* becomes -5.13 and is highly statistically significant. This implies that a treated firm with the average marginal corporate tax rate reduces its leverage by 5.13 percentage points in response to the 2003 tax cut.¹⁷ Our estimates in Model 2 of Table 3 also show that the impact of the tax cut disappears if the marginal corporate tax rate reaches 34.2%, which is effectively the top applicable rate for any firm in our sample.

5.4. Falsification tests

As pointed out by Angrist and Krueger (1999), a key identifying assumption for the DID methodology is that identification terms are zero in the absence of the intervention. Following Blouin, Raedy, and Shackelford (2011), we repeat the tests in the Sections 5.2 and 5.3 for other years when there was no change in personal tax rates. Specifically, we repeat the tests for three sample periods: from 2000 to 2002, from 2004 to 2006, and from 2006

¹⁵ Aiken and West (1991, p. 35) recommend the use of centered regressors to reduce collinearity between the interaction term and the variables that separately compose the interaction term.

¹⁶ We interact individual ownership with the marginal corporate tax rate after interest expense deductions. Appendix B explains why we think *Mtra* is the relevant tax rate for the test. The results change little if we use *Mtrb* instead.

¹⁷ This leverage reduction occurs in conjunction with higher dividend payments (Chetty and Saez, 2005), which would tend to raise leverage on their own.

Table 2

Change in leverage from 2002 to 2004.

The dependent variable is the change in book leverage from 2002 to 2004, measured in percentage points. Leverage is defined as the sum of long-term debt (DLT) and short-term debt (DLCC) divided by total assets (AT). *IndOwn* is the fraction of shares owned by individual investors, which is computed as one minus the fraction of shares owned by institutional investors at the most recent quarter end before the fiscal year end, obtained from Thomson Reuters Institutional Holdings (13F) Database. Cash flow (*CF*) is defined as operating income before depreciation (OIBDP) over total assets. *Momentum* is the firm's buy-and-hold return in the 12 months before the end of the fiscal year 2002. Market-to-book ratio (*MB*) is market value of equity (CSHO*PRCC_C) over book value of equity (CEQ). $\ln(\text{assets})$ is the logarithm of total assets. Collateralizable assets (*Col*) is the sum of inventories (INVT) and property, plant and equipment (PPENT) over total assets. Intangible assets (*Intang*) is intangible assets (INTAN) over total assets. Dividend payment (*Ddiv*) is equal to one if common dividends (DVC) is greater than zero, and zero otherwise. All variables in levels are measured at the end of 2002. Δ denotes the change in the value of each variable from 2002 to 2004. Standard errors are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

Variable	(1)	(2)
<i>IndOwn</i>	-1.68** (0.81)	-4.95*** (1.06)
<i>CF</i>		-9.87*** (1.30)
<i>Momentum</i>		-1.07*** (0.38)
<i>MB</i>		-0.67*** (0.13)
$\ln(\text{assets})$		-0.22 (0.16)
$\Delta \ln(\text{assets})$	5.74*** (0.56)	7.34*** (0.59)
ΔCol	15.01*** (2.63)	15.25*** (2.58)
ΔMB	0.74*** (0.11)	0.37*** (0.12)
ΔIntang	8.39*** (2.29)	6.71*** (2.25)
ΔCF	-12.63*** (1.56)	-20.64*** (1.78)
ΔDdiv	-1.70* (0.88)	-0.75 (0.87)
R-squared	0.067	0.101
N	3,051	3,051

to 2008. All variables in the hypothetical event periods are measured as they are for 2002 to 2004. Finding similar relations between the change in the debt ratio and *IndOwn* in years without tax rate reductions would raise questions about the implications of the results in Tables 2 and 3.

The results of this exercise are reported in Table 4. None of the coefficients of *IndOwn* differs significantly from zero. The interaction term *IndOwn* $\times$ *Mtra* is also insignificant except in 2006–2008 when its sign is the opposite of the coefficient in 2002–2004. The corporate tax rate carries a significant coefficient only in 2004–2006. But the coefficients on *Mtra* have large standard errors in all four time periods, and the estimated effects do not differ reliably across time. Overall, the findings in Table 4 support the assumption that individual ownership generally does not matter for firms' leverage changes. The results in Tables 2

Table 3

Change in leverage from 2002 to 2004 and marginal corporate tax rates.

The dependent variable is the change in book leverage from 2002 to 2004, measured in percentage points. Leverage is defined as the sum of long-term debt (DLT) and short-term debt (DLCC) divided by total assets (AT). Marginal tax rates (*Mtra*) are firms' marginal corporate tax rates after interest deductions as estimated by Blouin, Core, and Guay (2010). Both *IndOwn* and *Mtra* are recentered at their respective means. All other variables are defined in Table 2. All variables in levels are measured at the end of 2002. Δ denotes the change in the value of each variable from 2002 to 2004. Standard errors are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

Variable	(1)	(2)
<i>IndOwn</i> _{centered}	-3.11*** (0.91)	-5.13*** (1.09)
<i>IndOwn</i> _{centered} $\cdot$ <i>Mtra</i> _{centered}	27.01*** (7.45)	37.24*** (7.49)
<i>Mtra</i> _{centered}	-7.66*** (2.29)	2.82 (2.54)
<i>CF</i>		-11.60*** (1.42)
<i>Momentum</i>		-1.15*** (0.38)
<i>MB</i>		-0.61*** (0.13)
$\ln(\text{assets})$		-0.29* (0.17)
$\Delta \ln(\text{assets})$	6.17*** (0.57)	7.62*** (0.60)
ΔCol	14.56*** (2.63)	14.22*** (2.59)
ΔMB	0.69*** (0.11)	0.37*** (0.12)
ΔIntang	7.74*** (2.29)	6.12*** (2.25)
ΔCF	-13.67*** (1.59)	-21.49*** (1.78)
ΔDdiv	-1.32 (0.88)	-0.58 (0.87)
R-squared	0.073	0.109
N	3,034	3,034

and 3 identify the causal effect of the 2003 tax cut rather than some spurious relations.

5.5. Financial market access

In this subsection, we test another prediction of Eq. (10): The larger the slope of a firm's marginal cost function of debt, the smaller the impact of a tax cut on its leverage ratio. If a firm has poor capital market access, it probably is limited in its ability to revise leverage following the JGTRRA. We, therefore, separate firms into two groups on the basis of their equity or long-term debt issuances and repurchases in the year before the JGTRRA (2002). Specifically, we define a firm as financially unconstrained if

$$\frac{\text{Long-term debt issuances}}{\text{Total assets}} \geq 66\text{th percentile in 2002}$$

$$\text{or } \frac{\text{Long-term debt reductions}}{\text{Total assets}} \geq 66\text{th percentile in 2002}$$

$$\text{or } \frac{\text{Equity issuances}}{\text{Total assets}} \geq 66\text{th percentile in 2002}$$

$$\text{or } \frac{\text{Equity repurchases}}{\text{Total assets}} \geq 66\text{th percentile in 2002.}$$

Table 4

Falsification test.

The dependent variable is the change in book leverage during each time period, measured in percentage points. Leverage is defined as the sum of long-term debt (DLT) and short-term debt (DLCC) divided by total assets (AT). *IndOwn* is the fraction of shares owned by individual investors, which is computed as one minus the fraction of shares owned by institutional investors at the most recent quarter end before the fiscal year end, obtained from Thomson-Reuters Institutional Holdings (13F) Database. Both *IndOwn* and *Mtra* are recentered at their respective means. All other variables are defined in Tables 2 and 3. All variables in levels are measured at the end of the year preceding the estimation interval. Δ denotes the change in the value of each variable during the estimation interval. Standard errors are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

Variable	2000–2002		2004–2006		2006–2008	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>IndOwn</i> _{centered}	−0.74 (1.19)	−0.98 (1.24)	−0.74 (1.05)	−0.34 (1.07)	−0.98 (1.18)	−0.70 (1.20)
<i>IndOwn</i> _{centered} · <i>Mtra</i> _{centered}		9.44 (11.73)		−4.53 (7.41)		−17.79* (9.95)
<i>Mtra</i> _{centered}		−1.28 (4.03)		6.29** (2.87)		6.75 (4.31)
<i>CF</i>	−13.47*** (1.37)	−12.92*** (1.60)	−5.82*** (1.39)	−7.36*** (1.63)	−3.12* (1.70)	−3.91* (2.05)
<i>Momentum</i>	0.71*** (0.27)	0.73*** (0.27)	0.03 (0.35)	0.04 (0.35)	−1.75*** (0.64)	−1.65** (0.64)
<i>MB</i>	−0.07 (0.13)	−0.10 (0.13)	−0.21* (0.12)	−0.19 (0.17)	0.13 (0.17)	0.13 (0.17)
$\ln(\text{assets})$	0.50*** (0.17)	0.50*** (0.18)	−0.08 (0.17)	−0.16 (0.17)	0.50*** (0.19)	0.42** (0.20)
$\Delta \ln(\text{assets})$	2.79*** (0.62)	2.81*** (0.62)	6.22*** (0.67)	6.09*** (0.67)	3.32*** (0.83)	3.19*** (0.83)
ΔCol	20.85*** (2.64)	20.67*** (2.68)	18.97*** (2.94)	18.58*** (2.95)	14.46*** (3.32)	14.53*** (3.32)
ΔMB	0.06 (0.14)	0.03 (0.14)	0.06 (0.13)	0.06 (0.13)	0.19 (0.18)	0.19 (0.18)
ΔIntang	5.74** (2.24)	5.56** (2.29)	14.35*** (2.33)	14.11*** (2.34)	6.24** (2.92)	6.11** (2.92)
ΔCF	−14.78*** (1.56)	−14.45*** (1.57)	−17.35*** (1.96)	−17.62*** (1.96)	−19.91*** (2.04)	−19.85*** (2.05)
ΔDdiv	−3.05*** (1.11)	−3.05*** (1.11)	−1.63* (0.98)	−1.67* (0.98)	−1.92 (1.36)	−1.96 (1.36)
<i>R</i> -squared	0.092	0.089	0.094	0.095	0.062	0.064
<i>N</i>	2,536	2,504	2,577	2,573	2,154	2,152

Firms that do not meet any of these conditions are treated as financially constrained.¹⁸

Table 5 reports the results of estimating three regression specifications separately for the unconstrained and constrained subsamples. Comparing Columns 1–3 against Columns 4–6, financially unconstrained firms reduce their debt ratio by 5.79 (6.32) percentage points when *Mtra* is not controlled for (*Mtra* is at its mean). These numbers are about 20% larger than those in Tables 2 and 3 when the whole sample is used. In contrast, the tax cut has no significant effect on the firms that appear to have worse access to external capital market. In other words, financing conditions plausibly affected firms' response to the 2003 personal tax cut.¹⁹

Taken together, the results in Section 5 provide strong evidence that personal equity taxes do matter for firms' capital structure decisions. The 2003 cut in personal equity taxes leads firms to use significantly less debt. The effect is stronger for firms with lower marginal corporate tax and firms that are financially unconstrained.

6. Robustness check

In this section, we provide three robustness checks on the main results in Section 5.

¹⁸ van Binsbergen, Graham, and Yang (2010) partition their firms into financially constrained and unconstrained groups using the prior year's market access (as we do), but they further require firms to be nondistressed, that is, having a z-score in the top third of the sample. Imposing this additional condition on our data leaves us with relatively few unconstrained firms. Furthermore, we feel that capital market participation alone implies relatively easy access to external markets, on average.

¹⁹ The results in Table 5 are not robust to three alternative measures of financial constraints (size, dividend policy, and credit rating). Upon inspecting the correlations among the three variables, we found that dividend payments and credit ratings have large positive correlations

(footnote continued)

with the size-based proxy for financial constraint. By contrast, our unconstrained measure has only low correlations with the three alternative proxies. Because larger firms also have lower individual ownership, their response to the JGTRRA could be small on account of their ownership structures. In other words, the unconstrained firms defined by size, dividends, or public debt rating tend to have small reactions to the JGTRRA for reasons other than the marginal cost of funding. While we think recent market access could be a better measure of a firm's easiness to adjust its leverage, the lack of robustness indicates that the results in Table 5 should be viewed as suggestive, not definitive, evidence.

Table 5

Financial market access and firms' response to 2003 tax cut.

The dependent variable is the change in book leverage from 2002 to 2004, measured in percentage points. Firms with *Part*=1 had relatively high capital market activity in 2002. Leverage is defined as the book value of long-term debt (DLT) and short-term debt (DLCC) divided by total assets (AT). Both *IndOwn* and *Mtra* are recentered at their respective means. All other variables are defined in Tables 2 and 3. All variables in levels are measured at the end of 2002. Δ denotes the change in the value of each variable from 2002 to 2004. Standard errors are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

Variable	<i>Part</i> =1			<i>Part</i> =0		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>IndOwn</i> _{centered}	−2.51*** (0.96)	−5.79*** (1.24)	−6.32*** (1.28)	0.66 (1.49)	−1.94 (2.09)	−1.62 (2.11)
<i>IndOwn</i> _{centered} · <i>Mtra</i> _{centered}			46.84*** (9.00)			13.28 (13.26)
<i>Mtra</i> _{centered}			2.55 (3.07)			3.53 (4.37)
<i>CF</i>		−9.23*** (1.55)	−11.09*** (1.69)		−12.65*** (2.43)	−13.86*** (2.62)
<i>Momentum</i>		−1.27*** (0.45)	−1.39*** (0.45)		−0.46 (0.68)	−0.48 (0.68)
<i>MB</i>		−0.65*** (0.15)	−0.56*** (0.15)		−0.71** (0.31)	−0.68** (0.31)
<i>Ln(assets)</i>		−0.26 (0.19)	−0.33* (0.20)		−0.03 (0.31)	−0.07 (0.32)
$\Delta \text{Ln(assets)}$	5.97*** (0.68)	7.27*** (0.71)	7.60*** (0.71)	5.54*** (1.02)	8.04*** (1.09)	8.09*** (1.10)
ΔCol	13.92*** (3.18)	14.53*** (3.13)	13.45*** (3.12)	15.40*** (4.61)	15.76*** (4.50)	15.24*** (4.52)
ΔMB	0.70*** (0.13)	0.33** (0.14)	0.38*** (0.14)	0.70*** (0.20)	0.27 (0.21)	0.30 (0.21)
ΔIntang	7.38*** (2.71)	6.36** (2.67)	5.52** (2.66)	11.57*** (4.31)	7.07* (4.28)	7.12* (4.28)
ΔCF	−12.49*** (1.89)	−20.61*** (2.18)	−21.61*** (2.18)	−13.19*** (2.69)	−21.40*** (3.00)	−21.70*** (3.01)
ΔDdiv	−1.65 (1.02)	−0.82 (1.01)	−0.70 (1.01)	−1.38 (1.75)	−0.26 (1.72)	−0.23 (1.73)
<i>R-squared</i>	0.065	0.096	0.107	0.082	0.129	0.131
<i>N</i>	2,293	2,293	2,293	741	741	741

6.1. Partial adjustment model

In Section 5, we focus on firms' leverage change from 2002 to 2004 to study the impact of the 2003 tax cut. This approach is intuitive and concise and implicitly accounts for leverage adjustment cost by including several firm characteristics in 2002 in the regression. In this section, we use a partial adjustment model from the capital structure literature to capture firms' gradual adjustment to their optimal leverage ratios, and we incorporate the treatment effect of the 2003 tax cut into the partial adjustment model to study the magnitude of its impact.

The illustration of the partial adjustment model in this section follows Flannery and Rangan (2006). The optimal or target leverage has the form

$$Blev_{i,t+1}^* = \phi X_{i,t}, \quad (20)$$

where $Blev_{i,t+1}^*$ is firm *i*'s desired debt ratio at *t*+1 and $X_{i,t}$ is a vector of firm characteristics related to the costs and benefits of operating with various leverage ratios.

Firms optimally deviate from their target leverage ratios if adjustment costs are substantial. This effect is captured by the standard partial adjustment model

$$Blev_{i,t+1} - Blev_{i,t} = \lambda(Blev_{i,t+1}^* - Blev_{i,t}) + \epsilon_{i,t+1}. \quad (21)$$

Each year, the typical firm closes a proportion λ of the gap between its actual and its desired leverage levels. Substituting Eq. (20) into Eq. (21) and rearranging terms gives the estimable model

$$Blev_{i,t+1} = (\lambda\phi)X_{i,t} + (1-\lambda)Blev_{i,t} + \epsilon_{i,t+1}. \quad (22)$$

It is straightforward to incorporate a treatment effect (e.g., tax cut) into this model. Suppose a firm's optimal leverage ratio changes after the 2003 tax cut, and let *Post* be an indicator variable for the time of the tax cut (i.e., *Post*=1 if *t*+1≥2003, and *Post*=0 otherwise). Then Eq. (20) is modified to be

$$Blev_{i,t+1}^* = \phi X_{i,t} + \gamma \text{Post} + \delta \text{Post} \times D_{i,2002}, \quad (23)$$

where $D_{i,2002}$ includes variables that are proxies for firm *i*'s treatment status in 2003. Analogous to Eq. (19), $D_{i,2002}$ consists of $\text{IndOwn}_{i,2002,\text{centered}}$, $\text{IndOwn}_{i,2002,\text{centered}} \times \text{Mtra}_{i,2002,\text{centered}}$, and $\text{Mtra}_{i,2002,\text{centered}}$. Therefore, δ identifies the effect of the tax cut. γ measures any change in firms' optimal leverage after 2003 that is not due to the tax cut. Accordingly, Eq. (22) becomes

$$Blev_{i,t+1} = (\lambda\phi)X_{i,t} + (\lambda\gamma)\text{Post} + (\lambda\delta)\text{Post} \times D_{i,2002} + (1-\lambda)Blev_{i,t} + \epsilon_{i,t+1}, \quad (24)$$

Table 6

Partial adjustment model.

We estimate a partial adjustment model of book debt ratio:

$$\begin{aligned}
Blev_{i,t+1} = & \alpha + \beta_1 Blev_{i,t} + \beta_2 Post + \beta_3 Post \times IndOwn_{i,2002,centered} \\
& + \beta_4 Post \times IndOwn_{i,2002,centered} \times Mtra_{i,2002,centered} \\
& \times Mtra_{i,2002,centered} + \beta_5 Post \times Mtra_{i,2002,centered} \\
& + \beta_6 IndOwn_{i,t} + \beta_7 Ln(assets)_{i,t} + \beta_8 MB_{i,t} \\
& + \beta_9 Intang_{i,t} + \beta_{10} CF_{i,t} + \beta_{11} Col_{i,t} + \beta_{12} Ddiv_{i,t} + u_i + \epsilon_{i,t+1}
\end{aligned}$$

Book leverage (*Blev*) is defined as the sum of long-term debt (DLT) and short-term debt (DLCC) divided by total assets (AT), measured in percentage points. *Post* is a dummy variable equal to one if $t \geq 2002$ and zero otherwise. Both *IndOwn* and *Mtra* in 2002 are recentered at their respective means. All other variables are defined in Tables 2 and 3. The sample includes firms in Compustat from 1998 to 2009. All regressions include firm fixed effects. Column 2 is estimated using the Stata command *xtabond2* with the *nolevel* option. Independent variables from time 1 up to time $t-1$ are used as instruments for the difference equation. Standard errors are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

Variable	Fixed effects	Difference GMM
<i>Blev</i>	0.55*** (0.01)	0.65*** (0.01)
<i>Post</i>	-1.32*** (0.15)	-0.89** (0.37)
<i>Post</i> · <i>IndOwn</i> _{2002,centered}	-1.54*** (0.45)	-1.41* (0.78)
<i>Post</i> · <i>IndOwn</i> _{2002,centered} · <i>Mtra</i> _{2002,centered}	9.24** (3.61)	57.71*** (18.76)
<i>Post</i> · <i>Mtra</i> _{2002,centered}	-4.01*** (1.14)	-7.95*** (3.03)
<i>IndOwn</i>	-1.50*** (0.46)	-8.54*** (1.18)
<i>Ln(assets)</i>	1.09*** (0.14)	0.37 (0.43)
<i>Col</i>	3.61*** (0.75)	0.31 (2.54)
<i>MB</i>	-0.06** (0.03)	-0.11** (0.05)
<i>Intang</i>	4.54*** (0.67)	5.21*** (1.71)
<i>Profit</i>	-2.84*** (0.45)	3.68*** (0.92)
<i>Ddiv</i>	1.41*** (0.27)	1.34* (0.71)
<i>R-squared</i>	0.327	
<i>N</i>	28,329	24,769

δ can be obtained by dividing the coefficient of the interaction term $Post \times D_{i,2002}$ by one minus the coefficient of lag leverage $Blev_{i,t}$. The model to be estimated in this section is

$$\begin{aligned}
Blev_{i,t+1} = & \alpha + \beta_1 Blev_{i,t} + \beta_2 Post + \beta_3 Post \\
& \times IndOwn_{i,2002,centered} + \beta_4 Post \\
& \times IndOwn_{i,2002,centered} \times Mtra_{i,2002,centered} \\
& + \beta_5 Post \times Mtra_{i,2002,centered} + \beta_6 IndOwn_{i,t} \\
& + \beta_7 Ln(assets)_{i,t} + \beta_8 MB_{i,t} + \beta_9 Intang_{i,t} + \beta_{10} CF_{i,t} \\
& + \beta_{11} Col_{i,t} + \beta_{12} Ddiv_{i,t} + u_i + \epsilon_{i,t+1}, \quad (25)
\end{aligned}$$

where the control variables are defined as before and u_i is a firm fixed effect. We include the firm fixed effects in all the regressions because they are known to be important in the study of capital structure with panel data (Flannery

and Rangan, 2006; Lemmon, Roberts, and Zender, 2008). Eq. (25) is analogous to Eq. (19), and the interpretation of the coefficients β_3 to β_5 is the same as before except that the impact must be adjusted by a factor of $1/(1-\beta_1)$. For example, $\beta_3/(1-\beta_1)$ measures the long-run impact of the tax cut on a treated firm's optimal leverage when its marginal corporate tax rate is at its mean.

The first column of Table 6 reports the estimates of the above fixed effects model. The coefficients of $Post \times IndOwn_{i,2002}$, $Post \times Ind Own_{i,2002} \times Mtra_{i,2002}$, and $Post \times Mtra_{i,2002}$ are statistically significant and have the expected signs. However, it is well known in the dynamic panel regression that the fixed effects estimation of adjustment speed is biased because the lag dependent variable is correlated with the error term (Flannery and Rangan, 2006; Huang and Ritter, 2009). Therefore, to address this issue, we estimate the regression with a Difference GMM method, in which all independent variables from time 1 up to time $t-1$ are used as instruments for the difference equation.²⁰ Column 2 shows that the coefficient of $Post \times IndOwn_{2002}$ with *Mtra*₂₀₀₂ centered at its mean is -1.41%, which translates to a reduction of $0.0141/(1-0.65) = 4.03\%$ in leverage ratio. This number is smaller than but not statistically different from the estimate of 5.13% in Table 3. Therefore, accounting for firms' gradual leverage adjustment does not change our main results.

6.2. Tax status of institutional investors

In Section 5, we measure individual ownership as one minus the fraction of shares owned by institutional investors and use the individual ownership as a proxy for the probability that a firm is in the treatment group. However, individual ownership computed this way does not perfectly measure the fraction of investors affected by the 2003 tax cut. Chetty and Saez (2005) argue that this raw institutional ownership fails to take into account the tax status of institutional investors. For example, mutual funds include investments that are subject to personal taxes and investments that are not subject to personal taxes. Blouin, Raedy, and Shackelford (2011) note that "sometimes, the dividends and capital gains realized by mutual funds are taxed at the individual level. At other times, distributions to mutual funds are exempt because the shares are held in deferred tax accounts." It is not clear whether our results are biased because of this measurement error. The bias depends on how the individual ownership used in our regressions is correlated with the fraction of indirect individual holdings through mutual fund. If they are not

²⁰ We use the Stata command *xtabond2* written by Roodman (2009) for the estimation. The overidentifying restrictions condition is rejected by the Sargan Test, which raises concerns about the validity of the instrumental variables used in the estimation. Using System GMM does not solve this problem. Besides, some of the estimated coefficients of variables such as *Size* and *IndMedian* have unexpected signs. Studies such as Lemmon, Roberts, and Zender (2008) and Flannery and Hankins (2013) that use System GMM often have the same issue. Therefore, we choose to use the Difference GMM in this paper. We acknowledge that, however, our GMM results are not robust to all the choices of instruments (e.g., using difference or level of variables as instruments, number of lags) used in the estimation.

correlated, then our estimate is still unbiased (see Wooldridge, 2001, pp. 73–74).

To evaluate this measurement error issue, we follow Blouin, Raedy, and Shackelford (2011) by controlling for the fraction of shares owned by mutual funds in our regression.²¹ Institutional managers in 13F are classified by Thomson-Reuters into five types, with typecode 3 being mutual funds. However, starting from 1998 this classification is wrong due to a mapping error by Thomson that improperly classifies institutions in the first four categories into group 5, with the vast majority of investment advisers being mistakenly allocated to the “all others” group. Therefore, Chetty and Saez (2005) manually reclassify the group 5 institutions. We use their reclassification to correct for the errors in 13F.²² The mean and median of mutual fund ownership in 2002 are 9.0% and 10.6%, respectively. Table 7 reports the results after controlling for mutual funds ownership in the regressions. Without including the interaction terms of ownership with marginal corporate tax rates, the coefficient of *IndOwn* is –5.32%, which is a bit larger than the –4.95% in Table 2. After the interaction terms are included, the coefficient of *IndOwn* becomes –4.51%, which is a bit smaller than the –5.13% in Table 3. Overall, Table 7 shows that our previous findings are not driven by the measurement error of individual ownership.

6.3. Using abnormal returns to identify firms with marginal individual investors

In this subsection, we provide evidence for the impact of the 2003 tax cut using an alternative proxy to identify the marginal investor: firms' abnormal stock returns associated with news about the 2003 tax cut. The tax cut would increase the value of future payouts to individual investors. Accordingly, firms that had positive abnormal returns during the periods when the tax cut became more likely to be implemented are more likely to have marginal individual investors. Amromin, Harrison, and Sharpe (2008) identify two periods during which positive news about the 2003 tax cut reached the market: January 3–9, 2003 and May 14–28, 2003. The newspaper coverage about the tax cut initially spiked in January 3–9, 2003. May 14, 2003 witnessed the first major breakthrough in Senate negotiations, and the president signed the bill on May 28. Firms' abnormal returns during these two periods could indicate the identity of their marginal investors: The higher the abnormal return, the more likely the marginal investor is a retail investor.

²¹ Chetty and Saez (2005) use a different method to deal with this issue. They isolate a subset of institutional investors that are definitely not affected by the reform and use firms whose largest institutional owner is a nonaffected entity as control group and all other firms as the treatment group. They define nonaffected entities in the Thomson database classified as insurance companies (Type 2) and those classified as other (Type 5) whose names indicate that they are a pension fund, nonprofit institution, government agency, or non-financial corporation. They have 8,065 observations in their treatment group versus 394 observations in the control group.

²² The reclassification is from Professor Raj Chetty's website: <http://obs.rc.fas.harvard.edu/chetty/chettypaperschronology.html>.

Table 7

Mutual fund ownership and change in debt ratio.

Dependent variable is the change in book leverage from 2002 to 2004, measured in percentage points. Leverage is defined as the sum of long-term debt (DLT) and short-term debt (DLCC) divided by total assets (AT). *MF* is the percentage of shares owned by mutual funds, which includes type 3 institution investors in Thomson Reuters Institutional Holdings (13F) Database and the type 3 institution investors from reclassification of type 5 investors in 13F by Chetty and Saez (2005). *IndOwn*, *Mtra*, and *MF* are recentered at their respective means. All other variables are defined in Tables 2 and 3. All variables in levels are measured at the end 2002. Δ denotes the change in the value of each variable from 2002 to 2004. Standard errors are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

Variable	(1)	(2)
<i>IndOwn</i> _{centered}	–5.32*** (1.34)	–4.51*** (1.36)
<i>IndOwn</i> _{centered} · <i>Mtra</i> _{centered}		18.77* (10.56)
<i>MF</i> _{centered}	–1.77 (3.80)	4.66 (4.12)
<i>MF</i> _{centered} · <i>Mtra</i> _{centered}		–87.54** (34.37)
<i>Mtra</i> _{centered}		2.25 (2.55)
<i>CF</i>	–9.84*** (1.30)	–11.70*** (1.42)
<i>Momentum</i>	–1.07*** (0.38)	–1.14*** (0.38)
<i>MB</i>	–0.66*** (0.13)	–0.61*** (0.13)
<i>Ln(assets)</i>	–0.21 (0.17)	–0.32* (0.17)
$\Delta \text{Ln(assets)}$	7.34*** (0.59)	7.57*** (0.60)
ΔCol	15.27*** (2.58)	14.17*** (2.58)
ΔMb	0.37*** (0.12)	0.37*** (0.12)
ΔIntang	6.66*** (2.26)	6.31*** (2.26)
ΔCF	–20.63*** (1.78)	–21.44*** (1.78)
ΔDdiv	–0.74 (0.87)	–0.56 (0.87)
<i>R</i> -squared	0.101	0.111
<i>N</i>	3,051	3,034

Our event windows and measures of abnormal returns follow Amromin, Harrison, and Sharpe (2008). For each firm we estimate the simple market model

$$r_{i,t} = \alpha_i + \beta_i r_{M,t} + \epsilon_{i,t}, \quad (26)$$

where $r_{M,t}$ is the value-weighted CRSP daily market returns. The estimation window is a 12-month period that straddles the event windows (July–December 2002 and July–December 2003). The results are not sensitive to a choice of the longer estimation period. We then compute the cumulative abnormal returns (CAR) in the event window by subtracting the expected return based on Eq. (26) from the actual return and scaling by the standard deviation of the residuals in Eq. (26) to account for firms' idiosyncratic risk:

$$\text{CAR}_i = \frac{\sum_t (r_{i,t} - \hat{\alpha}_i - \hat{\beta}_i r_{M,t})}{\sigma_i \sqrt{N}}, \quad (27)$$

Table 8

Change in leverage from 2002 to 2004 and firms' abnormal returns.

The dependent variable is the change in leverage from 2002 to 2004, measured in percentage points. Leverage is defined as the book value of long-term debt (DLT) and short-term debt (DLCC) divided by total assets (AT). CAR^1 and CAR^2 are firms' abnormal stock returns in event windows January 3–9, 2003 and May 14–28, 2003, respectively. Both CAR^1 and CAR^2 are scaled by the product of the standard deviation of the error term in the estimation window and the square root of event days in each event window. *Treat* is a dummy variable equal to one if both CAR^1 and CAR^2 are greater than the number in each column header, and zero otherwise. All other control variables are defined in Tables 2 and 3. Δ denotes the change in the value of each variable from 2002 to 2004. Standard errors are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

Variable	<i>Treat</i> = 1 when both CAR^1 and CAR^2 are greater than				
	0.2	0.4	0.6	0.8	1.0
<i>Treat</i>	0.21 (0.67)	−1.62* (0.84)	−1.88* (1.06)	−3.78*** (1.32)	−5.21*** (1.73)
<i>CF</i>	−9.42*** (1.31)	−9.64*** (1.31)	−9.64*** (1.31)	−9.71*** (1.31)	−9.71*** (1.30)
<i>Momentum</i>	−1.14*** (0.38)	−1.17*** (0.38)	−1.16*** (0.38)	−1.16*** (0.38)	−1.15*** (0.38)
<i>MB</i>	−0.62*** (0.13)	−0.63*** (0.13)	−0.63*** (0.13)	−0.63*** (0.13)	−0.63*** (0.13)
$\ln(\text{assets})$	0.25* (0.13)	0.24* (0.13)	0.24* (0.13)	0.23* (0.13)	0.24* (0.13)
$\Delta \ln(\text{assets})$	7.57*** (0.59)	7.58*** (0.59)	7.58*** (0.59)	7.55*** (0.59)	7.55*** (0.59)
ΔCol	15.89*** (2.59)	15.88*** (2.59)	15.89*** (2.59)	15.86*** (2.58)	15.87*** (2.58)
ΔMb	0.37*** (0.12)	0.38*** (0.12)	0.37*** (0.12)	0.38*** (0.12)	0.38*** (0.12)
$\Delta Intang$	7.23*** (2.26)	7.42*** (2.26)	7.44*** (2.26)	7.46*** (2.26)	7.40*** (2.25)
ΔCF	−20.64*** (1.79)	−20.62*** (1.79)	−20.58*** (1.79)	−20.54*** (1.79)	−20.52*** (1.79)
$\Delta Ddiv$	−0.65 (0.87)	−0.66 (0.87)	−0.64 (0.87)	−0.66 (0.87)	−0.66 (0.87)
<i>R-squared</i>	0.095	0.096	0.095	0.097	0.097
<i>N</i>	3,051	3,051	3,051	3,051	3,051
Number of <i>Treat</i> = 1	537	347	237	175	132

where σ_i is the standard deviation of the error term $\epsilon_{i,t}$ in Eq. (27) and N is the number of event days in the event windows.²³ We estimate the average CAR for the two event windows used in Amromin, Harrison, and Sharpe (2008). We denote the January CAR as CAR^1 and the May CAR as CAR^2 .

Other news (unrelated to the tax cut legislation) arrived during these two periods, giving some firms abnormal returns even if they were unaffected by the tax cut. The largest CAR_i 's are most likely to indicate firms with individuals as their marginal investors and, hence, are most likely to be affected by the tax cut. We, therefore, define a variety of treated firm samples by including only those with sufficiently positive abnormal returns during the two event periods. We create an indicator variable *Treat* for the treatment status of a firm: *Treat* = 1 if a firm has large positive abnormal returns in both periods and *Treat* = 0 otherwise.²⁴ *Treat*, therefore, has the same meaning as d_i in Section 3. Because we do not have a theoretically motivated cutoff for this classification, we estimate

the results for a series of models featuring various cutoffs for the definition of *Treat*.

Table 8 reports the estimated impact of the 2003 tax cut on firms' leverage based on our new classification of treated firms and control firms. The *Treat* dummy is statistically significant in most of our specifications. Moreover, the coefficient of the *Treat* dummy becomes more negative as we require a higher abnormal return for a firm to be classified as treated. This could reflect the fact that a low cutoff could misclassify some control firms as treated firms. When we require a firm to have abnormal returns at least as large as the standard deviation of returns in both periods to be a treated firm (the last column), the treated firms on average reduce their leverage ratios 5.21 percentage points more than control firms do. Overall, Table 8 confirms our results in previous sections that the debt ratios of firms affected by the 2003 tax cut decrease significantly more than those of unaffected firms.

We show the positive relation between *Treat* and individual ownership in Table 9, which reports a Probit regression of the dummy variable *Treat* on individual ownership and the control variables used in previous tables. The marginal effects at means are reported in Table 9. Firms with higher individual ownership do tend to have higher abnormal returns. The first two columns show that a firm with 100% individual ownership is 7% or 18% more likely to be a treated firm than a firm with zero

²³ We obtain similar results without adjusting for the idiosyncratic risk.

²⁴ We also estimate our models by classifying a firm as treated if it has large positive abnormal returns in either of the two event windows. We obtain quantitatively similar results.

Table 9

Probit regression of treatment on individual ownership.

This table reports the marginal effects at means from a probit regression. The dependent variable is the *Treat* dummy variable defined in Table 8. Book leverage is defined as the sum of long-term debt (DLT) and short-term debt (DLCC) divided by total assets (AT). CAR^1 and CAR^2 are firms' abnormal stock returns in event windows January 3–9, 2003 and May 14–28, 2003, respectively. Both CAR^1 and CAR^2 are scaled by the product of the standard deviation of the error term in the estimation window and the square root of event days in each event window. *Treat* is a dummy variable equal to one if both CAR^1 and CAR^2 are greater than the number in each column header, and zero otherwise. All other control variables are defined in Tables 2 and 3. Standard errors are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

Variable	<i>Treat</i> = 1 when both CAR^1 and CAR^2 are greater than							
	0.4 (1)	0.4 (2)	0.6 (3)	0.6 (4)	0.8 (5)	0.8 (6)	1.0 (7)	1.0 (8)
<i>IndOwn</i>	0.18*** (0.02)	0.07*** (0.02)	0.15*** (0.01)	0.04** (0.02)	0.13*** (0.01)	0.03** (0.01)	0.10*** (0.01)	0.02*** (0.01)
<i>Ln(assets)</i>		–0.00 (0.00)		–0.00 (0.00)		–0.00 (0.00)		–0.00 (0.00)
<i>MB</i>		–0.01*** (0.00)		–0.01*** (0.00)		–0.01*** (0.00)		–0.00*** (0.00)
<i>Intang</i>		–0.01 (0.03)		–0.00 (0.02)		–0.00 (0.01)		0.01 (0.01)
<i>Col</i>		–0.07*** (0.02)		–0.06*** (0.02)		–0.03*** (0.01)		–0.01* (0.01)
<i>CF</i>		–0.06*** (0.02)		–0.05*** (0.01)		–0.03*** (0.01)		–0.02*** (0.01)
<i>Ddiv</i>		–0.02 (0.01)		–0.00 (0.01)		–0.00 (0.01)		0.01 (0.00)
<i>Momentum</i>		–0.02* (0.01)		–0.01 (0.01)		–0.00 (0.01)		–0.00 (0.00)
Pseudo R-squared	0.050	0.054	0.071	0.078	0.108	0.099	0.151	0.135
<i>N</i>	3,306	3,089	3,306	3,089	3,306	3,089	3,306	3,089
Number of <i>Treat</i> = 1	347	347	237	237	175	175	132	132

Table 10

Change in leverage from 2002 to 2004, individual ownership, and abnormal returns.

The dependent variable is the change in book leverage from 2002 to 2004, measured in percentage points. Leverage is defined as the sum of long-term debt (DLT) and short-term debt (DLCC) divided by total assets (AT). CAR^1 and CAR^2 are firms' abnormal stock returns in event windows January 3–9, 2003 and May 14–28, 2003, respectively. Both CAR^1 and CAR^2 are scaled by the product of the standard deviation of the error term in the estimation window and the square root of event days in each event window. *Treat* is a dummy variable equal to one if both CAR^1 and CAR^2 are greater than the number in each column header, and zero otherwise. All other control variables are defined in Tables 2 and 3. Δ denotes the change in the value of each variable from 2002 to 2004. Standard errors are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

Variable	<i>Treat</i> = 1 when both CAR^1 and CAR^2 are greater than				
	0.2	0.4	0.6	0.8	1.0
<i>Treat</i>	0.34 (0.67)	–1.42* (0.84)	–1.69 (1.06)	–3.53*** (1.32)	–4.86*** (1.73)
<i>IndOwn</i>	–4.97*** (1.07)	–4.85*** (1.06)	–4.88*** (1.06)	–4.83*** (1.06)	–4.81*** (1.06)
<i>CF</i>	–9.81*** (1.31)	–10.02*** (1.31)	–10.03*** (1.31)	–10.09*** (1.30)	–10.09*** (1.30)
<i>Momentum</i>	–1.06*** (0.38)	–1.09*** (0.38)	–1.08*** (0.38)	–1.08*** (0.38)	–1.08*** (0.38)
<i>MB</i>	–0.67*** (0.13)	–0.68*** (0.13)	–0.68*** (0.13)	–0.68*** (0.13)	–0.68*** (0.13)
<i>Ln(assets)</i>	–0.22 (0.16)	–0.22 (0.16)	–0.22 (0.16)	–0.22 (0.16)	–0.22 (0.16)
Δ <i>Ln(assets)</i>	7.33*** (0.59)	7.35*** (0.59)	7.34*** (0.59)	7.32*** (0.59)	7.32*** (0.59)
Δ <i>Col</i>	15.24*** (2.58)	15.25*** (2.58)	15.25*** (2.58)	15.23*** (2.58)	15.24*** (2.58)
Δ <i>Mb</i>	0.37*** (0.12)	0.37*** (0.12)	0.37*** (0.12)	0.37*** (0.12)	0.38*** (0.12)
Δ <i>Intang</i>	6.69*** (2.25)	6.88*** (2.25)	6.90*** (2.25)	6.93*** (2.25)	6.87*** (2.25)
Δ <i>CF</i>	–20.63*** (1.78)	–20.61*** (1.78)	–20.58*** (1.78)	–20.54*** (1.78)	–20.52*** (1.78)
Δ <i>Ddiv</i>	–0.75 (0.87)	–0.76 (0.87)	–0.74 (0.87)	–0.76 (0.87)	–0.76 (0.87)
R-squared	0.101	0.102	0.102	0.103	0.103
<i>N</i>	3,051	3,051	3,051	3,051	3,051
Number of <i>Treat</i> = 1	537	347	237	175	132

individual ownership, with or without control variables in the regression. In some specifications, the pseudo R -squared exceeds 10% even when individual ownership is the only independent variable in the regression.

Finally, in Table 10 we include both individual ownership and the *Treat* dummy in the leverage change regressions. In all specifications, the coefficient of *IndOwn* is significantly negative. As we increase the cutoffs on the abnormal returns for a firm to be defined as treated, the *Treat* dummy becomes significant as well. Although neither ownership nor returns proxies perfectly for the treatment status of firms in 2003, the heterogeneity in firms' leverage adjustment in 2003 caused by these two variables enables us to identify the causal impact of the tax cut.

7. Conclusion

The Jobs and Growth Tax Relief Reconciliation Act of 2003 substantially reduced the personal tax rates on dividends and capital gains while leaving the tax rate on interest income unchanged. This event, therefore, provides a clean setting to test the effect of personal taxes on the cost of equity and optimal capital structure. Using a difference-in-differences method, we estimate that the 2003 tax cut reduced affected firms' book leverage by about 5 percentage points on average. Because the tax cut applied only to individual investors, its effect depended on each firm's marginal investor's type. We use the firm's individual ownership to proxy for the probability that the marginal investor is an individual and, thus, the firm is affected by the tax cut. The advantage of this method is that it mitigates any confounding change in firms' leverage in 2003 and enables us to interpret the results as causal effects instead of simple correlation. The limitation of this method is that individual ownership is not a perfect identification variable. By repeating our regressions in years with no tax change, we attempt to exclude alternative explanations and the possibility of spurious relations. We also use firms' abnormal returns in two event windows in early 2003 as an alternative proxy for the treatment status of firms in 2003. We find that firms with

high positive abnormal returns (those we classify as treated firms) reduce their leverage significantly more than firms with low or negative returns. Finally, providing further evidence for the causal explanation of the tax cut, we show that the effects of the tax cut are stronger for firms with lower marginal corporate tax rates and for firms that have worse capital market access. Overall, we find strong evidence that personal tax is an important determinant of firms' optimal leverage.

Appendix A. Sample selection

Book leverage is defined as the book value of long-term debt (DLT) and short-term debt (DLCC) divided by total assets (AT). *InstOwn* is the fraction of shares owned by institutional investors, obtained from Thomson-Reuters Institutional Holdings (13F) Database. Market-to-book ratio (*MB*) is market value of equity ($CSHO \times PRCC_C$) over book value of equity (*CEQ*). Cash flow (*CF*) is defined as operating income before depreciation (OIBDP) over assets. *Momentum* is the firm's buy-and-hold return in the 12 months before the end of the fiscal year. The sample period is 1998–2009.

Filters	Observations left
Compustat firms with non-missing <i>InstOwn</i>	61,519
Book leverage is not missing and between zero and one	60,548
Non-financial, non-utility firms	47,155
Firms have consecutive ownership data from 2002–2004	34,158
Winsorization	Changes made
Replace <i>InstOwn</i> = 0 if <i>InstOwn</i> is missing in 13F, but the firm is in CRSP	384
Replace <i>InstOwn</i> = 1 if <i>InstOwn</i> > 1	1,162
Replace <i>MB</i> = 10 if <i>MB</i> > 10	1,739
Replace <i>MB</i> = 0.1 if <i>MB</i> < 0.1	1,070
Replace <i>CF</i> = 1 if <i>CF</i> > 1	2
Replace <i>CF</i> = -1 if <i>CF</i> < -1	473
Replace <i>Momentum</i> = 10 if <i>Momentum</i> > 10	36

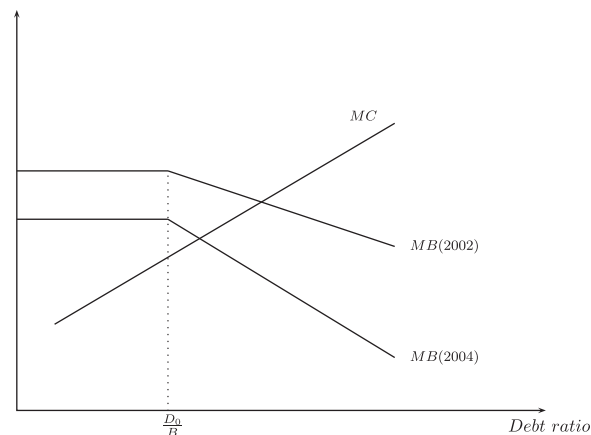


Fig. B1. Marginal cost (*MC*) and downward-sloping marginal benefit (*MB*) of debt.

Appendix B. Comparative static results with a sloping corporate tax schedule

The derivation of Eqs. (5), (9), and (10) assume that the corporate tax rate is unaffected by a firm's amount of outstanding debt, as in Fig. 1. In reality, however, the marginal tax rate should decline (at least weakly) as a firm increases its debt ratio. Fig. B1 illustrates a more realistic marginal benefit schedule for a firm's debt financing. The marginal corporate tax rate remains constant with debt variations up to D_0/B . Beyond this point, more debt

financing reduces the firm's tax rate. The impact of a tax rate change that shifts the MB curve in Fig. B1, therefore, depends on the curve's slope. For debt ratios above D_0/B , τ_c is a function of D , with $\tau'_c(D) \leq 0$.²⁵ Eq. (7) therefore becomes

$$V_L = V_U + \int_0^D \left[1 - \frac{(1-\tau_c(D'))(1-\tau_e)}{1-\tau_p} \right] dD' - \frac{B \times C\left(\frac{rD}{B}\right)}{(1-\tau_p)r}. \quad (28)$$

The benefit of tax-sheltered interest payments is captured by the first term and the cost of outstanding debt is measured by the second term. Differentiating Eq. (28) with respect to the debt level gives a first order condition that is similar to Eq. (8):

$$1 - \frac{(1-\tau_c(D^*))(1-\tau_e)}{1-\tau_p} = \frac{(\alpha + \beta \frac{rD^*}{B})}{1-\tau_p}. \quad (29)$$

Unlike the case of a constant τ_c , we cannot solve Eq. (29) for an explicit optimal debt level. Nevertheless, implicitly differentiating Eq. (29) with respect to τ_e yields the following impact of the equity tax rate on optimal debt:

$$\frac{\partial D^*}{\partial \tau_e} = \frac{1-\tau_c(D^*)}{\beta r - B(1-\tau_e) \frac{d\tau_c}{dD^*}}. \quad (30)$$

When firms are operating on the flat marginal benefit line, $d\tau_c/dD^* = 0$ and Eq. (30) reduces to Eq. (10). Because $d\tau_c/dD^*$ is otherwise negative, the denominator of Eq. (30) is positive and firms with higher τ_c should have smaller debt adjustments following the JGTTRA, as in Eq. (10).²⁶ Eq. (30) also specifies that the change in leverage associated with a change in τ_e depends on the firm's initial corporate tax rate. In our empirical estimations, we therefore use the firm's marginal tax rate after interest deductions, which we call $Mtra$.

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²⁵ Although the figure shows $\tau_c(D)$ as a smooth curve for simplicity, the US tax code would make the actual tax rate decline in steps with higher leverage.

²⁶ The prediction that JGTTRA induces larger changes at firms with lower τ_c is unambiguous in most situations. However, the result is ambiguous if we compare a relatively high corporate tax rate firm with an equilibrium on the horizontal part of the marginal benefit curve against a relatively low tax rate firm with an equilibrium on the downward-sloping part of the marginal benefit curve. In this case, the predicted change in leverage following JGTTRA depends on the relative slopes of the marginal cost and marginal benefit curves, as well as on their initial τ_c values. For these firms, the effect of τ_c on the change in optimal leverage must be determined empirically.